Use Case Specification

Hasnahena Platform

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Actor interactions and system behaviors
for all platform workflows

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Document Control

Item	Details
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Ver.	Date	Author	Changes
1.0	March 22, 2026	Md Saif Ahmed	Initial release

1 Actors

1.1 Primary Actors

Actor	Description
Member	Community member who deposits, borrows, donates, and acts as guarantor
Worker	Staff member who processes physical/digital fund transfers
Admin	Administrator with full system privileges for approvals, configuration, and oversight

1.2 External Actors

Actor	Description
Notification Service	External SMS/email service for sending alerts
NID Verification API	Porichoy API for National ID verification

2 Use Case Summary

ID	Name	Summary
UC-01	Register Member	Collect member profile, NID, nominee; queue for admin review
UC-02	Approve Member	Admin approves/rejects registration and triggers credential notification
UC-03	Request Deposit	Member submits deposit request and receives status tracking
UC-04	Process Deposit	Worker confirms receiving funds and marks transaction processed
UC-05	Confirm Deposit	Member verifies amount and confirms final posting
UC-06	Raise Dispute	Member disputes mismatched amount; system freezes and alerts admin
UC-07	Apply for Loan	Eligible member submits loan request with purpose and guarantor details
UC-08	Guarantor Approval	Selected guarantor accepts/rejects support obligation
UC-09	Approve Loan	Admin approves/rejects request after policy checks
UC-10	Disburse Loan	Worker disburses approved loan; member confirms receipt
UC-11	Record Installment	Worker/Admin records repayment and updates loan state

ID	Name	Summary
UC-12	Default Settlement	Admin applies settlement waterfall according to policy
UC-13	Donate to Fund	Member/worker records donation into selected fund bucket

3 Detailed Use Case Specifications

3.1 UC-01: Register Member

Field	Details
Use Case ID	UC-01
Name	Register Member
Primary Actor	Visitor (prospective member)
Stakeholders	Admin, Organization
Trigger	Visitor accesses registration form

Preconditions:

- Registration form is publicly accessible
- Visitor has valid NID and nominee information

Main Flow:

1. Visitor opens the public registration form
2. System displays form with required fields
3. Visitor enters personal details (name, phone, email, NID, address)
4. Visitor enters nominee details (name, NID, relationship)
5. Visitor uploads required documents (photo, NID copy)
6. Visitor accepts terms and submits form
7. System validates input and stores registration
8. System queues registration for admin review
9. System displays confirmation message

Alternative Flows:

- **A1:** Validation fails → System highlights errors, visitor corrects and resubmits
- **A2:** Duplicate NID detected → System rejects with "already registered" message

Postconditions:

- Registration record created with PENDING status
- Admin notification triggered

3.2 UC-07: Apply for Loan

Field	Details
Use Case ID	UC-07
Name	Apply for Loan
Primary Actor	Member
Stakeholders	Admin, Guarantor(s), Worker
Trigger	Member clicks "Apply for Loan" button

Preconditions:

- Member account is active
- Minimum deposit requirement is satisfied (8,000 BDT)
- No existing active loan (unless policy allows multiple)

Main Flow:

1. Member opens loan application form
2. System displays maximum eligible amount based on caps
3. Member enters requested amount and purpose
4. System validates requested amount against eligibility
5. If amount exceeds threshold, member selects guarantor(s)
6. System stores loan request with PENDING_GUARANTOR status
7. System sends notification to selected guarantor(s)

Alternative Flows:

- **A1:** Requested amount exceeds allowed cap → System rejects with reason and displays max amount
- **A2:** Amount below threshold → No guarantor required, status set to PENDING_APPROVAL
- **A3:** Selected guarantor is ineligible → System prompts to select different guarantor

Postconditions:

- Loan request is either pending admin review (if complete) or awaiting guarantor/member action
- Audit log entry created

3.3 UC-06: Raise Dispute

Field	Details
Use Case ID	UC-06
Name	Raise Dispute
Primary Actor	Member
Stakeholders	Admin, Worker
Trigger	Member identifies incorrect amount on processed transaction

Preconditions:

- Transaction is in PROCESSED status
- Transaction has not yet been confirmed or disputed

Main Flow:

1. Member views transaction in dashboard
2. Member clicks "Dispute" button
3. System prompts for dispute reason and correct amount
4. Member submits dispute
5. System changes transaction status to DISPUTED
6. System freezes transaction (prevents further state changes)
7. System sends alert to Admin

Postconditions:

- Transaction status is DISPUTED
- Admin notification sent
- Audit log records dispute action

3.4 UC-12: Default Settlement

Field	Details
Use Case ID	UC-12
Name	Default Settlement
Primary Actor	Admin
Stakeholders	Member (borrower), Guarantor(s), Organization
Trigger	Admin initiates settlement for defaulted loan

Preconditions:

- Loan is in DEFAULTED status
- Grace period has expired

Main Flow:

1. Admin reviews defaulted loan details
2. Admin initiates settlement waterfall
3. System calculates outstanding balance
4. System deducts from borrower's deposit account
5. If insufficient, system deducts from guarantor(s) deposits proportionally
6. If still insufficient, system flags as "Bad Debt" for charity fund write-off
7. System updates loan status to SETTLED
8. System sends notifications to affected parties

Postconditions:

- Loan marked as SETTLED
- All deductions recorded as transactions
- Affected accounts updated
- Complete audit trail recorded

4 Use Case Diagram

The visual use case diagram is maintained separately and includes:

- All actors and their relationships
- All use cases with «include» and «extend» relationships
- System boundary demarcation

Key relationships:

- UC-03 (Request Deposit) «include» UC-05 (Confirm Deposit)
- UC-07 (Apply for Loan) «include» UC-08 (Guarantor Approval)
- UC-05 (Confirm Deposit) «extend» UC-06 (Raise Dispute)